

Section 3 - Practice Problems

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Chapter 3.1: Sequences and Their Limits

Exercise 3.1.7 Let $x_n := \frac{1}{\ln(n+1)}$ for $n \in \mathbb{N}$.

(a) Use the definition of limits to show that $\lim_{n \rightarrow \infty} (x_n) = 0$.

Solution:

Let $\epsilon > 0$ be arbitrary. Since $\ln$ is an increasing function, we know that $\ln(n) \leq \ln(n+1)$ and thus $\frac{1}{\ln(n)} \geq \frac{1}{\ln(n+1)}$. Let $N > e^{\frac{1}{\epsilon}}$. We therefore know that $\forall n \geq N$:

$$\begin{aligned} |x_n - 0| &= |x_n| \\ &= \left| \frac{1}{\ln(n+1)} \right| \\ &= \frac{1}{\ln(n+1)} \\ &\leq \frac{1}{\ln(n)} \\ &\leq \frac{1}{\ln(N)} \\ &< \frac{1}{\ln(e^{\frac{1}{\epsilon}})} \\ &= \frac{1}{\frac{1}{\epsilon}} \\ &= \epsilon \end{aligned}$$

Therefore, by definition of limits, $\lim_{n \rightarrow \infty} (x_n) = 0$.

- (b) Find a specific value of N as required in the definition of limits for each of the following ϵ -challenges:

(I) $\epsilon = \frac{1}{2}$.

Solution:

From our prior definition of N , we can select any natural number greater than:

$$\begin{aligned} e^{\frac{1}{\epsilon}} &= e^{\frac{1}{\frac{1}{2}}} \\ &= e^2 \\ &\approx 7.3890561 \end{aligned}$$

Thus, we can choose $N = 8$ to satisfy this ϵ -challenge.

(II) $\epsilon = \frac{1}{10}$

Solution:

For the same reason outlined earlier, we must select an N greater than:

$$\begin{aligned} e^{\frac{1}{\epsilon}} &= e^{\frac{1}{\frac{1}{10}}} \\ &= e^{10} \\ &\approx 22026.46 \end{aligned}$$

Thus, we can choose $N = 22027$ to satisfy this ϵ -challenge.

Exercise 3.1.8

- (a) Prove that $\lim_{n \rightarrow \infty} (x_n) = 0$ if and only if $\lim_{n \rightarrow \infty} (|x_n|) = 0$.

Solution:

Proof.

(I) Prove forward direction ($\lim_{n \rightarrow \infty} (x_n) = 0 \implies \lim_{n \rightarrow \infty} (|x_n|) = 0$)

Let $\epsilon > 0$ be arbitrary.

We know that the sequence $(|x_n|)$ converges to zero if for any such ϵ , there exists some $N \in \mathbb{N}$ such that $\forall n \geq N$, $||x_n| - 0| < \epsilon$.

Let's briefly turn our attention to the sequence (x_n) . Since x_n converges to zero, then we know that there exists an $N_0 \in \mathbb{N}$ such that $\forall n \geq N_0$, $|x_n - 0| < \epsilon$. However, considering that $||x_n|| = |x_n|$, we necessarily have, $\forall n \geq N_0$:

$$\begin{aligned} ||x_n| - 0| &= ||x_n|| \\ &= |x_n| \\ &= |x_n - 0| \\ &< \epsilon \end{aligned}$$

Thus, since $||x_n| - 0| < \epsilon$ for arbitrary $\epsilon > 0$ so long as $n \geq N_0$, by definition of convergence, the sequence $(|x_n|)$ converges to zero and thus $\lim_{n \rightarrow \infty} (x_n) = 0 \implies \lim_{n \rightarrow \infty} (|x_n|) = 0$.

(II) Prove reverse direction ($\lim_{n \rightarrow \infty} (|x_n|) = 0 \implies \lim_{n \rightarrow \infty} (x_n) = 0$).

Once again, let $\epsilon > 0$ be arbitrary.

We know that the sequence (x_n) converges to zero if, for any such ϵ , there exists an $N \in \mathbb{N}$ such that $\forall n \geq N$, $|x_n - 0| < \epsilon$.

Using the same idea as before, we know that there exists an $N_0 \in \mathbb{N}$ such that $\forall n \geq N$, $||x_n| - 0| < \epsilon$ since $(|x_n|)$ converges to zero. Thus, $\forall n \geq N_0$:

$$\begin{aligned} |x_n - 0| &= |x_n| \\ &= ||x_n|| \\ &= ||x_n| - 0| \\ &< \epsilon \end{aligned}$$

Thus, since $|x_n - 0| < \epsilon$ for arbitrary $\epsilon > 0$ so long as $n \geq N_0$, by definition of convergence, the sequence (x_n) converges to zero and thus $\lim_{n \rightarrow \infty} (|x_n|) = 0 \implies \lim_{n \rightarrow \infty} (x_n) = 0$.

Therefore, since:

$$\begin{aligned} \lim_{n \rightarrow \infty} (x_n) = 0 &\implies \lim_{n \rightarrow \infty} (|x_n|) = 0 \\ &\text{and} \\ \lim_{n \rightarrow \infty} (|x_n|) = 0 &\implies \lim_{n \rightarrow \infty} (x_n) = 0 \end{aligned}$$

$$\lim_{n \rightarrow \infty} (x_n) = 0 \iff \lim_{n \rightarrow \infty} (|x_n|) = 0 \quad \square$$

(b) Give an example to show that the convergence of $|x_n|$ need not imply the convergence of (x_n) .

Solution:

The sequence $x_n = (-1)^n$ is a very simply example which illustrates this fact. Clearly, $x_n = (-1)^n$ is divergent, however $|x_n| = 1 \forall n \in \mathbb{N}$ and thus $|x_n|$ is clearly convergent to 1. Thus, $|x_n|$ being convergent does not necessarily imply that x_n is.

Exercise 3.1.14 Let $b \in \mathbb{R}$ satisfy $0 < b < 1$. Show that $\lim_{n \rightarrow \infty} (nb^n) = 0$.

Solution:

Exercise 3.1.17 Show that $\lim_{n \rightarrow \infty} (\frac{2^n}{n!}) = 0$. [Hint: If $n \geq 3$, then $0 < \frac{2^n}{n!} < 2 \cdot (\frac{2}{3})^{n-2}$]

Solution:

Let $\epsilon > 0$ be arbitrary. To prove that $\lim_{n \rightarrow \infty} \frac{2^n}{n!} = 0$, we must show that there exists some $N \in \mathbb{N}$ such that $n \geq N \implies |\frac{2^n}{n!} - 0| < \epsilon$. To start, we begin by setting $N_1 = 3$. Thus, using the hint given in the question, $\forall n \geq N_1$, we have:

$$\begin{aligned} \left| \frac{2^n}{n!} - 0 \right| &= \left| \frac{2^n}{n!} \right| \\ &\leq \left| 2 \cdot \left(\frac{2}{3} \right)^{n-2} \right| \\ &= \left| \frac{9}{2} \left(\frac{2}{3} \right)^n \right| \\ &= \frac{9}{2} \left(\frac{2}{3} \right)^n \end{aligned}$$

Now let's consider the sequence $(\frac{2}{3})^n$. Our goal is to prove that this sequence converges to zero, because if it does, then we can make that term arbitrarily small in our previous

expression and our proof that $\lim_{n \rightarrow \infty} (\frac{2^n}{N!}) = 0$ is complete. Let $N_2 > \log_{\frac{2}{3}}(\frac{2\epsilon}{9})$, we know that $\forall n \geq N_2 \implies$:

$$\begin{aligned} |(\frac{2}{3})^n - 0| &= |(\frac{2}{3})^n| \\ &\geq |(\frac{2}{3})^{N_2}| < |(\frac{2}{3})^{\log_{\frac{2}{3}}(\frac{2\epsilon}{9})}| \\ &= |\frac{2\epsilon}{9}| \\ &= \frac{2\epsilon}{9} \end{aligned}$$

Therefore, by definition of convergence, we know that $\lim_{n \rightarrow \infty} (\frac{2}{3})^n = 0$. Thus, by letting $N = \max\{N_1, N_2\}$, then $n \geq N \implies$:

$$\begin{aligned} |\frac{2^n}{n!} - 0| &= |\frac{2^n}{n!}| \\ &\leq |2 \cdot (\frac{2}{3})^{n-2}| \\ &= |\frac{9}{2} \cdot (\frac{2}{3})^n| \\ &= \frac{9}{2} \cdot (\frac{2}{3})^n \\ &< \frac{9}{2} \cdot \frac{2\epsilon}{9} \\ &= \epsilon \end{aligned}$$

Hence, by definition of convergence, $\lim_{n \rightarrow \infty} (\frac{2^n}{n!}) = 0$.

Exercise 3.1.18 If $\lim_{n \rightarrow \infty} (x_n) = x > 0$, show that there exists a $K \in \mathbb{N}$ such that if $n \geq K$, then $\frac{1}{2}x < x_n < 2x$.

Solution:

Let $\epsilon = \frac{1}{2}x$ (which we can do since $x > 0$). By definition of convergence, we know that there exists a $K \in \mathbb{N}$ such that $\forall n \geq K$:

$$|x_n - x| < \epsilon = \frac{1}{2}x$$

However, notice that:

$$\begin{aligned} |x_n - x| &< \frac{1}{2}x \implies \\ -\frac{1}{2}x &< x_n - x < \frac{1}{2}x \implies \\ -\frac{1}{2}x + x &< x_n - x + x < \frac{1}{2}x + x \implies \\ \frac{1}{2}x &< x_n < \frac{3}{2}x \implies \\ \frac{1}{2}x &< x_n < 2x \end{aligned}$$

Thus, $\forall n \geq K, \frac{1}{2}x < x_n < 2x$.

Chapter 3.2: Limit Theorems

Exercise 3.2.7 If (b_n) is a bounded sequence and $\lim_{n \rightarrow \infty} (a_n) = 0$, prove that $\lim_{n \rightarrow \infty} (a_n \cdot b_n) = 0$. Explain why Theorem 3.2.3 *cannot* be used.

Solution:

Proof. Let $\epsilon > 0$ be arbitrary. Since b_n is given to be bounded, we know that there exists an $M \in \mathbb{R}, M > 0$ such that $|b_n| \leq M \ \forall n \in \mathbb{N}$. Furthermore, because a_n converges to 0, we know that there exists an $N \in \mathbb{N}$ such that $n \geq N \implies |a_n - 0| = |a_n| < \frac{\epsilon}{M}$. Thus, $n \geq N \implies$:

$$\begin{aligned} |a_n \cdot b_n - 0| &= |a_n \cdot b_n| \\ &\leq M \cdot |a_n| \\ &< M \cdot \frac{\epsilon}{M} \\ &< \epsilon \end{aligned}$$

Thus, by definition of convergence, $\lim_{n \rightarrow \infty} (a_n \cdot b_n) = 0$. □

We could not have simply invoked Theorem 3.2.3 to prove this since (b_n) is simply given to be bounded, not convergent.

Exercise 3.2.8 Explain why the result in equation (3) before theorem 3.2.4 *cannot* be used to evaluate the limit of the sequence $\lim_{n \rightarrow \infty} ((1 + \frac{1}{n})^n)$.

Solution:

Equation 3 states that:

$$\lim_{n \rightarrow \infty} (a_n^k) = (\lim_{n \rightarrow \infty} (a_n))^k, \quad \forall k \in \mathbb{N}$$

We cannot use this to evaluate the limit of the sequence $\lim_{n \rightarrow \infty} ((1 + \frac{1}{n})^n)$ since the exponent here is not a constant k but rather a limiting variable. If we were to attempt to apply this law, we would get that $\lim_{n \rightarrow \infty} ((1 + \frac{1}{n})^n) = 1$, which as was determined earlier, would be incorrect.

Exercise 3.2.11 Determine the following limits:

(a) $\lim_{n \rightarrow \infty} ((3\sqrt{n})^{\frac{1}{2n}})$.

Exercise 3.2.16 Apply theorem 3.2.11 to the following sequences, where a, b satisfy $0 < a < 1, b > 1$.

(a) (a^n) .

Solution:

Theorem 3.2.11 states that for a sequence (x_n) of real numbers, that if the sequence $(\frac{x_{n+1}}{x_n})$ exists and is less than 1, then (x_n) converges and $\lim_{n \rightarrow \infty} (x_n) = 0$

For the sequence a^n , we know that:

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(\frac{a^{n+1}}{a^n} \right) &= \lim_{n \rightarrow \infty} (a) \\ &= a \end{aligned}$$

And since $0 < a < 1$, we know that the sequence a^n converges and $\lim_{n \rightarrow \infty} a^n = 0$.

(b) $(\frac{b^n}{2^n})$.

Solution:

$$\lim_{n \rightarrow \infty} \left(\frac{b^{n+1}/2^{n+1}}{b^n/2^n} \right) = \lim_{n \rightarrow \infty} \left(\frac{b}{2} \right)$$

Therefore, $\forall b > 2$, the sequence diverges, and for all $1 < b < 2$ the sequence converges to 0. If $b = 2$, we get that:

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(\frac{b^n}{2^n} \right) &= \lim_{n \rightarrow \infty} \left(\frac{2^n}{2^n} \right) \\ &= \lim_{n \rightarrow \infty} (1) \\ &= 1 \end{aligned}$$

Therefore, when $b = 2$, the sequence converges to 1.

(c) $(\frac{n}{b^n})$.

Solution:

$$\begin{aligned}\lim_{n \rightarrow \infty} \left(\frac{n + 1/b^{n+1}}{n/b^n} \right) &= \lim_{n \rightarrow \infty} \left(\frac{n + 1}{b^{n+1}} \cdot \frac{b^n}{n} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{b^n(n + 1)}{b^{n+1}(n)} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{n + 1}{b(n)} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{1}{b} + \frac{1}{b(n)} \right) \\ &= \frac{1}{b}\end{aligned}$$

Therefore, since $b > 1 \implies \frac{1}{b} < 1$, we know that $\lim_{n \rightarrow \infty} (\frac{n+1/b^{n+1}}{n/b^n}) < 1$ and thus $\frac{n}{b^n}$ converges to zero.

(d) $(\frac{2^{3n}}{3^{2n}})$.

Solution:

$$\begin{aligned}\lim_{n \rightarrow \infty} \left(\frac{2^{3(n+1)}/3^{2(n+1)}}{2^{3n}/3^{2n}} \right) &= \lim_{n \rightarrow \infty} \left(\frac{(2^3 \cdot 2^{3n})/(3^2 \cdot 3^{2n})}{(2^{3n})/(3^{2n})} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{2^3}{3^2} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{8}{9} \right) \\ &= \frac{8}{9}\end{aligned}$$

Therefore, since $\lim_{n \rightarrow \infty} \left(\frac{2^{3(n+1)}/3^{2(n+1)}}{2^{3n}/3^{2n}} \right) = \frac{8}{9} < 1$, the sequence $\left(\frac{2^{3n}}{3^{2n}} \right)$ converges to zero.

Exercise 3.2.17

(a) Give an example of a convergent sequence (x_n) of positive numbers with $\lim_{n \rightarrow \infty} \left(\frac{x_{n+1}}{x_n} \right)$.

Solution:

The sequence $x_n = 1$ is trivially convergent to 1 and has the property that:

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(\frac{x_{n+1}}{x_n} \right) &= \lim_{n \rightarrow \infty} \left(\frac{1}{1} \right) \\ &= \lim_{n \rightarrow \infty} (1) \\ &= 1 \end{aligned}$$

(b) Give an example of a divergent sequence with this property. (Thus, this property cannot be used as a test for convergence).

Solution:

The sequence $x_n = n$ is an example of a sequence which is clearly divergent (since (x_n) is unbounded), with the property that:

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(\frac{x_{n+1}}{x_n} \right) &= \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right) \\ &= \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right) \\ &= \lim_{n \rightarrow \infty} (1) + \lim_{n \rightarrow \infty} \left(\frac{1}{n} \right) \\ &= 1 + 0 \\ &= 1 \end{aligned}$$

Exercise 3.2.19 Discuss the convergence of each of the following sequences, where a, b satisfy $0 < a < 1, b > 1$.

(a) $(n^2 a^n)$.

Solution:

$$\begin{aligned}\lim_{n \rightarrow \infty} \left(\frac{(n+1)^2 / a^{n+1}}{n^2 / a^n} \right) &= \lim_{n \rightarrow \infty} \left(\frac{(n+1)^2}{a^{n+1}} \cdot \frac{a^n}{n^2} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{(n+1)^2 (a^n)}{a^{n+1} (n)^2} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{(n+1)^2}{a(n)^2} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{n^2 + 2n + 1}{a(n)^2} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{1}{a} + \frac{2}{a(n)} + \frac{1}{a(n)^2} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{1}{a} \right) + \lim_{n \rightarrow \infty} \left(\frac{2}{a(n)} \right) + \lim_{n \rightarrow \infty} \left(\frac{1}{a(n)^2} \right) \\ &= \frac{1}{a}\end{aligned}$$

Since $0 < a < 1 \implies \frac{1}{a} > 1$, $\lim_{n \rightarrow \infty} \left(\frac{(n+1)^2 / a^{n+1}}{n^2 / a^n} \right) > 1$ and thus the sequence is divergent.

(b) $\frac{b^n}{n^2}$.

Solution:

$$\begin{aligned}
\lim_{n \rightarrow \infty} \left(\frac{b^{n+1}/(n+1)^2}{b^n/n^2} \right) &= \lim_{n \rightarrow \infty} \left(\frac{b^{n+1}}{(n+1)^2} \cdot \frac{n^2}{b^n} \right) \\
&= \lim_{n \rightarrow \infty} \left(\frac{b^{n+1}(n)^2}{(n+1)^2 b^n} \right) \\
&= \lim_{n \rightarrow \infty} \left(\frac{b(n)^2}{(n+1)^2} \right) \\
&= \lim_{n \rightarrow \infty} (b) \cdot \lim_{n \rightarrow \infty} \left(\frac{n^2}{(n+1)^2} \right) \\
&= b \cdot 1 \\
&= b
\end{aligned}$$

Therefore, since $b > 1$, $\lim_{n \rightarrow \infty} \left(\frac{b^{n+1}/(n+1)^2}{b^n/n^2} \right) > 1$ and thus the sequence is divergent.

(c) $\frac{b^n}{n!}$

Solution:

$$\begin{aligned}
\lim_{n \rightarrow \infty} \left(\frac{b^{n+1}/(n+1)!}{b^n/n!} \right) &= \lim_{n \rightarrow \infty} \left(\frac{b^{n+1}}{(n+1)!} \cdot \frac{n!}{b^n} \right) \\
&= \lim_{n \rightarrow \infty} \left(\frac{b^{n+1}n!}{b^n(n+1)!} \right) \\
&= \lim_{n \rightarrow \infty} \left(\frac{(b)n!}{(n+1)!} \right) \\
&= \lim_{n \rightarrow \infty} \left(\frac{(b)n!}{(n+1)n!} \right) \\
&= \lim_{n \rightarrow \infty} \left(\frac{b}{n+1} \right) \\
&= 0
\end{aligned}$$

Therefore, since $\lim_{n \rightarrow \infty} \frac{b^{n+1}/(n+1)!}{b^n/n!} = 0 < 1$, the sequence is convergent.

(d) $\frac{n!}{n^n}$

Solution:

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \left(\frac{(n+1)!/(n+1)^{n+1}}{n!/n^n} \right) &= \lim_{n \rightarrow \infty} \left(\frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} \right) \\
 &= \lim_{n \rightarrow \infty} \left(\frac{(n+1)!n^n}{n!(n+1)^{n+1}} \right) \\
 &= \lim_{n \rightarrow \infty} \left(\frac{(n+1)n!n^n}{n!(n+1)^{n+1}} \right) \\
 &= \lim_{n \rightarrow \infty} \left(\frac{(n+1)n^n}{(n+1)^{n+1}} \right) \\
 &= \lim_{n \rightarrow \infty} \left(\frac{n^n}{n+1^n} \right) \\
 &= \lim_{n \rightarrow \infty} \left(\left(\frac{n}{n+1} \right)^n \right) \\
 &= \lim_{n \rightarrow \infty} \left(\left(1 - \frac{1}{n+1} \right)^n \right) \\
 &< \lim_{n \rightarrow \infty} (1)^n \\
 &= 1
 \end{aligned}$$

Therefore, since $\lim_{n \rightarrow \infty} \left(\frac{(n+1)!/(n+1)^{n+1}}{n!/n^n} \right) < 1$, the sequence is convergent.

Exercise 3.2.20 Let (x_n) be a sequence of positive numbers such that $\lim_{n \rightarrow \infty} ((x_n)^{\frac{1}{n}}) = L < 1$. Show that there exists a number r with $0 < r < 1$ such that $0 < x_n < r^n$ for all sufficiently large $n \in \mathbb{N}$. Use this to show that $\lim_{n \rightarrow \infty} (x_n) = 0$.

Solution:

Let $\epsilon \in (0, 1 - L)$ and let $r = L + \epsilon$ (clearly then $0 < r < 1$). By definition of convergence, since the sequence $\lim_{n \rightarrow \infty} ((x_n)^{\frac{1}{n}})$ converges to L , we know that there exists an $N \in \mathbb{N}$ such that $\forall n \geq N$:

$$|x_n^{\frac{1}{n}} - L| < \epsilon$$

But that means:

$$\begin{aligned}
(x_n)^{\frac{1}{n}} - L < \epsilon &\implies \\
(x_n)^{\frac{1}{n}} < \epsilon + L &\implies \\
((x_n)^{\frac{1}{n}})^n < (\epsilon + L)^n &\implies \\
x_n < (\epsilon + L)^n &\implies \\
x_n < r^n
\end{aligned}$$

Therefore, since x_n is given to be positive, we have $0 < x_n < r^n$ for all $n \geq N$. Since $0 < r < 1$, we know that $\lim_{n \rightarrow \infty} (r^n) = 0$. Thus, by squeeze theorem, since $\lim_{n \rightarrow \infty} (0) = \lim_{n \rightarrow \infty} (r^n) = 0$, and $0 < x_n < r^n$ for sufficiently large n , $\lim_{n \rightarrow \infty} (x_n) = 0$.

Exercise 3.2.21

- (a) Give an example of a convergent sequence (x_n) of positive numbers such that $\lim_{n \rightarrow \infty} \left((x_n)^{\frac{1}{n}} \right) = 1$.

Solution:

The sequence $x_n = 1$ trivially converges to 1 with the property that:

$$\begin{aligned}
\lim_{n \rightarrow \infty} \left((x_n)^{\frac{1}{n}} \right) &= \lim_{n \rightarrow \infty} \left((1)^{\frac{1}{n}} \right) \\
&= \lim_{n \rightarrow \infty} (1) \\
&= 1
\end{aligned}$$

- (b) Give an example of a divergent sequence (x_n) of positive numbers with $\lim_{n \rightarrow \infty} \left((x_n)^{\frac{1}{n}} \right)$. (Thus, this property cannot be used as a test for convergence).

Solution:

The sequence $(x_n) = n$ is clearly divergent (since it is unbounded), with the property that:

$$\begin{aligned}\lim_{n \rightarrow \infty} \left((x_n)^{\frac{1}{n}} \right) &= \lim_{n \rightarrow \infty} (n)^{\frac{1}{n}} \\ &= 1\end{aligned}$$

Exercise 3.2.24 Show that if (x_n) , (y_n) , and (z_n) are convergent sequences, then the sequence (w_n) defined by $w_n := \text{mid}\{x_n, y_n, z_n\}$ is convergent.

Solution:

Solution:

Chapter 3.3: Monotone Sequences

Exercise 3.3.3 Let $x_1 \geq 2$ and $x_{n+1} := 1 + \sqrt{x_n - 1}$, $\forall n \in \mathbb{N}, n > 1$. Show that (x_n) converges and find the limit.

Solution:

To show that (x_n) is convergent, we can use the monotone convergence theorem if we are able to prove that the sequence is bounded and monotone. Let's first start with boundedness. Conjecturing that $x_n \geq 2, \forall n \in \mathbb{N}$, let $P(n)$ be the statement $P(n) : 2 \leq x_n$.

(I) Show that (x_n) is lowerbounded.

(I.i) Show the base case holds ($P(1)$ is true).

LHS:

$$= x_1$$

RHS:

$$= 2$$

Since we are given that $x_1 \geq 2$, we have $\text{LHS} \geq \text{RHS}$ and thus $P(1)$ is true.

(I.ii) Formulate induction hypothesis (State $P(k)$).

Letting $k \in \mathbb{N}$ be arbitrary, we can formulate $P(k)$ as:

$$P(k) : 2 \leq x_k$$

(I.iii) Prove $P(k) \implies P(k+1)$.

$$\begin{aligned} x_{k+1} &= 1 + \sqrt{x_k - 1} \\ &\geq 1 + \sqrt{2 - 1} \\ &= 1 + 1 \\ &= 2 \end{aligned}$$

Hence, $P(k) \implies P(k+1)$ for arbitrary $k \in \mathbb{N}$.

Thus, since $P(1)$ is true, and for arbitrary $k \in \mathbb{N}$, $P(k) \implies P(k+1)$, by principle of mathematical induction, $P(n)$ is true $\forall n \in \mathbb{N}$ and thus (x_n) is lowerbounded by 2.

(II) Show that (x_n) is monotonously decreasing.

Selecting $x_1 = 5$ and computing $x_2 = 1 + \sqrt{x_1 - 1} = 1 + \sqrt{4} = 3$, I conjecture that the sequence (x_n) is monotonously decreasing and thus that $x_{n+1} \leq x_n \forall n \in \mathbb{N}$. Let $P(n)$ be the statement $P(n) : x_{n+1} \leq x_n$. We can rigorously show this to be true for all choices of x_1 using induction in conjunction with the fact that $x_1 \geq 2 \implies x_1 - 1 \geq 1 \implies x_1 - 1 \geq \sqrt{x_1 - 1}$.

(II.i) Show the base case holds ($P(1)$ is true).

LHS:	RHS:
$= x_1$	$= x_2$
$= x_1$	$= 1 + \sqrt{x_1 - 1}$
$= x_1$	$\leq 1 + x_1 - 1$
$= x_1$	$= x_1$

(II.ii) Formulate induction hypothesis (State $P(k)$).

Letting $k \in \mathbb{N}$ be arbitrary, we can formulate $P(k)$ as:

$$P(k) : x_{k+1} \leq x_k$$

(II.iii) Prove $P(k) \implies P(k+1)$

$$\begin{aligned}
 x_{k+2} &= 1 + \sqrt{x_{k+1} - 1} \\
 &\leq 1 + x_{k+1} - 1 \\
 &= x_{k+1}
 \end{aligned}$$

Hence, $P(k) \implies P(k+1)$ for arbitrary $k \in \mathbb{N}$.

Thus, since $P(1)$ is true, and for arbitrary $k \in \mathbb{N}$, $P(k) \implies P(k+1)$, by principle of mathematical induction, $P(n)$ is true $\forall n \in \mathbb{N}$, and thus we know that the sequence (x_n) is monotonously decreasing.

Therefore, since (x_n) is both bounded below and monotonously decreasing by monotone convergence theorem, (x_n) converges. To find the limit, let $L = \lim_{n \rightarrow \infty} (x_n)$. Then we know:

$$\begin{aligned}
L &= \lim_{n \rightarrow \infty} (x_n) \implies \\
L &= \lim_{n \rightarrow \infty} \left(1 + \sqrt{x_{n-1} - 1} \right) \implies \\
L &= 1 + \sqrt{L - 1} \implies \\
L - 1 &= \sqrt{L - 1} \implies \\
(L - 1)^2 &= L - 1 \implies \\
L^2 - 2L + 1 &= L - 1 \implies \\
L^2 - 3L + 2 &= 0 \implies \\
(L - 2)(L - 1) &= 0
\end{aligned}$$

Thus, since we know that 2 is a lower bound of the sequence (x_n) , we know that $L = 2$ and thus $\lim_{n \rightarrow \infty} (x_n) = 2$.

Exercise 3.3.5 Let $y_1 := \sqrt{p}$ where $p > 0$ and $y_{n+1} = \sqrt{p + y_n}$, $\forall n \in \mathbb{N}$. Show that (y_n) converges and find the limit. [Hint: One upper bound is $1 + 2\sqrt{p}$].

Solution:

To prove the convergence of (y_n) , we first demonstrate that (y_n) is both upperbounded and increasing. Therefore, let $P_1(n)$ and $P_2(n)$ be the statements $P_1(n) : y_n \leq 1 + 2\sqrt{p}$ and $P_2(n) : y_n \leq y_{n+1}$ respectively. We will prove that both $P_1(n)$ and $P_2(n)$ hold $\forall n \in \mathbb{N}$ by induction.

(I) Prove (y_n) is upperbounded.

(I.i) Show the base case holds ($P_1(1)$ is true).

LHS:

$$\begin{aligned}
y_1 \\
= \sqrt{p}
\end{aligned}$$

RHS:

$$\begin{aligned}
1 + 2\sqrt{p} \\
= 1 + 2\sqrt{p}
\end{aligned}$$

Therefore, $P_1(1)$ is true.

(I.ii) Formulate induction hypothesis (State $P_1(k)$).

Letting $k \in \mathbb{N}$ be arbitrary, assume that the statement:

$$P(k) : y_k \leq 1 + 2\sqrt{p}$$

is true.

(I.iii) Prove $P_1(k) \implies P_1(k+1)$.

$$\begin{aligned} y_{k+1} &= \sqrt{p + y_k} \implies \\ (y_{k+1})^2 &= p + y_k \implies \\ (y_{k+1})^2 &\leq 1 + 2\sqrt{p} + p \implies \\ (y_{k+1})^2 &\leq (1 + \sqrt{p})^2 \implies \\ y_{k+1} &\leq 1 + \sqrt{p} \implies \\ y_{k+1} &\leq 1 + 2\sqrt{p} \end{aligned}$$

Hence, $P_1(k) \implies P_1(k+1)$ for arbitrary $k \in \mathbb{N}$.

Thus, since $P_1(1)$ is true, and for arbitrary $k \in \mathbb{N}$, $P_1(k) \implies P_1(k+1)$, by principle of mathematical induction, $P_1(n)$ is true $\forall n \in \mathbb{N}$, and thus we know that the sequence (y_n) is upper bounded by $1 + 2\sqrt{p}$.

(II) Show that (y_n) is monotonously increasing.

(II.i) Show the base case holds ($P_2(1)$ is true).

LHS	RHS:
$= y_1$	$= y_2$
$= \sqrt{p}$	$= \sqrt{p + y_1}$
$= \sqrt{p}$	$= \sqrt{p + \sqrt{p}}$
$= \sqrt{p}$	$\geq \sqrt{p}$

Thus, since $\text{LHS} \leq \text{RHS}$, $P_2(1)$ is true.

(II.ii) Formulate inductive hypothesis (State $P_2(k)$).

Letting $k \in \mathbb{N}$ be arbitrary, assume that the statement:

$$P_2(k) : y_k \leq y_{k+1}$$

is true.

(II.iii) Prove that $P_2(k) \implies P_2(k+1)$.

LHS:

$$\begin{aligned} y_{k+1} &= \sqrt{p + y_k} \\ &= \sqrt{p + y_k} \end{aligned}$$

RHS:

$$\begin{aligned} y_{k+2} &= \sqrt{p + y_{k+1}} \\ &\geq \sqrt{p + y_k} \end{aligned}$$

Hence, $P_2(k) \implies P_2(k+1)$ for arbitrary $k \in \mathbb{N}$.

Thus, since $P_2(2)$ is true, and for arbitrary $k \in \mathbb{N}$, $P_2(k) \implies P_2(k+1)$, by principle of mathematical induction, $P_2(n)$ is true $\forall n \in \mathbb{N}$, and thus we know that the sequence is monotonously increasing.

Therefore, since the sequence is upperbounded and monotonously increasing, by the monotone convergence theorem, (y_n) converges.

Let $L = \lim_{n \rightarrow \infty} (y_n)$. We therefore know that:

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} (y_n) = \lim_{n \rightarrow \infty} (y_{n+1}) \implies \\ L &= \lim_{n \rightarrow \infty} (\sqrt{p + y_n}) \implies \\ L &= \sqrt{p + L} \implies \\ L^2 &= p + L \implies \\ L^2 - L - p &= 0 \implies \\ L &= \frac{1 \pm \sqrt{1 - 4(1)(p)}}{2} \implies \\ L &= \frac{1 \pm \sqrt{1 + 4p}}{2} \end{aligned}$$

For all $p > 0$, we know that $1 + 4p > 1$ and thus $\frac{1 - \sqrt{1 + 4p}}{2} < 0$ for all p . Since the sequence is increasing, and $y_1 > 0$, we know that $L > 0$ as well and thus $L = \lim_{n \rightarrow \infty} (y_n) = \frac{1 + \sqrt{1 + 4p}}{2}$.

Exercise 3.3.8 Let (a_n) be an increasing sequence, (b_n) be a decreasing sequence, and assume that $a_n \leq b_n$, $\forall n \in \mathbb{N}$. Show that $\lim_{n \rightarrow \infty} (a_n) \leq \lim_{n \rightarrow \infty} (b_n)$, and thereby deduce the Nested Interval Property from the Monotone Convergence Theorem.

Solution:

(I) Prove $\lim_{n \rightarrow \infty} (a_n) \leq \lim_{n \rightarrow \infty} (b_n)$.

Proof. We know that (b_n) is a decreasing sequence and lower bounded by any term (such as a_1) in a_n . Therefore, by the Monotone Convergence Theorem, we know that $\lim_{n \rightarrow \infty} (b_n)$ exists. Likewise, we know $\lim_{n \rightarrow \infty} (a_n)$ exists, since (a_n) is an increasing sequence and bounded above by any term in (b_n) . Thus, since (a_n) and (b_n) are both convergent sequences satisfying $a_n \leq b_n$ $\forall n \in \mathbb{N}$, we can invoke Theorem 3.2.5 to conclude that $\lim_{n \rightarrow \infty} (a_n) \leq \lim_{n \rightarrow \infty} (b_n)$. $\square$

(II) Prove the Nested Interval Property using the Monotone Convergence Theorem.

Proof. Let $I_1 \supseteq I_2 \supseteq \dots \supseteq I_n \supseteq \dots$ be a sequence of non-empty, nested, and closed intervals where $I_n = [a_n, b_n]$. We know, for the sequences to be nested, that $a_n \leq a_{n+1}$ and $b_n \geq b_{n+1}$, $\forall n \in \mathbb{N}$ (see Exercise 2.5.6 for proof), implying that the sequence (a_n) and (b_n) are increasing and decreasing sequences respectively. Furthermore, for each I_n to be non-empty, we also know that $a_n \leq b_n$. Thus, using our prior findings, we know $\lim_{n \rightarrow \infty} (a_n)$ and $\lim_{n \rightarrow \infty} (b_n)$ both exist with $\lim_{n \rightarrow \infty} (a_n) \leq \lim_{n \rightarrow \infty} (b_n)$. Now consider $\bigcap_{n=0}^{\infty} I_n$. Since each set is nested, we know that $\bigcap_{n=0}^{\infty} I_n = [\lim_{n \rightarrow \infty} (a_n), \lim_{n \rightarrow \infty} (b_n)]$. However, since $\lim_{n \rightarrow \infty} (a_n) \leq \lim_{n \rightarrow \infty} (b_n)$, we know that the closed interval $[\lim_{n \rightarrow \infty} (a_n), \lim_{n \rightarrow \infty} (b_n)]$ is non-empty and hence we have derived the Nested Interval Property from the Monotone Convergence Theorem. $\square$

Exercise 3.3.9 Let A be an infinite subset of $\mathbb{R}$ that is bounded above and let $u := \sup(A)$. Show that there exists an increasing sequence (x_n) with $x_n \in A$ $\forall n \in \mathbb{N}$ such that $\lim_{n \rightarrow \infty} (x_n) = u$.

Solution:

Proof. If $u \in A$, then trivially we know that the sequence $(x_n) = u$ is an increasing sequence (since $x_n \leq x_{n+1}$, $\forall n \in \mathbb{N}$) that converges to u . If $u \notin A$, then we can rely on the definition of the supremum to construct x_n . Consider the sequence $\epsilon_n = \frac{1}{n}$. By definition of supremum, we know that $\exists a_n, a_{n+1} \in A$ such that $u - \epsilon_n < a_n$ and $u - \epsilon_{n+1} < a_{n+1}$. Using this notion, we can

construct an increasing (x_n) such that $x_1 \in (a_1, a_2)$, $x_2 \in (a_2, a_3)$, $\dots$, $x_n \in (a_n, a_{n+1})$, $\dots$. To prove that (x_n) converges to u , consider any $\epsilon > 0$. By the Archimedean principle, we know that there exists an $N \in \mathbb{N}$ such that $\epsilon_N = \frac{1}{N} < \epsilon$. This also clearly means that $\forall n \geq N, \epsilon_n = \frac{1}{n} < \epsilon$. Thus, $\forall x_n, n \geq N$, we have that:

$$\begin{aligned} x_n \in (a_n, a_{n+1}) &\implies \\ a_n < x_n < a_{n+1} &\implies \\ u - \epsilon_n < x_n &\implies \\ u - x_n < \epsilon_n &\implies \\ u - x_n < \epsilon &\implies \\ |u - x_n| < \epsilon \end{aligned}$$

Therefore, by definition of convergence, the sequence (x_n) converges and therefore, for all infinite subsets of $\mathbb{R}$ which are bounded above, there always exists a sequence contained within the set which converges to its supremum. $\square$

Exercise 3.3.10 Establish the convergence or divergence of the sequence $(y_n) := \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n}$, $\forall n \in \mathbb{N}$.

Solution:

I conjecture that (y_n) is a convergent sequence and will attempt to prove this using the monotone convergence theorem by showing that (y_n) is both upperbounded and increasing.

(I) Prove (y_n) is bounded above.

$$\begin{aligned} y_n &= \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} \\ &\leq \frac{1}{n+1} + \frac{1}{n+1} + \dots + \frac{1}{n+1} \\ &= \frac{n}{n+1} \\ &= 1 - \frac{1}{n+1} \\ &< 1 \end{aligned}$$

Thus, the sequence (y_n) is bounded above by 1.

(II) Prove (y_n) is monotonously increasing.

We can likewise prove that (y_n) is increasing simply through algebraic manipulation

$$\begin{aligned}
 y_{n+1} &= \frac{1}{(n+1)+1} + \frac{1}{(n+1)+2} + \dots + \frac{1}{(n+1)+n-1} + \frac{1}{(n+1)+n} + \frac{1}{(n+1)+n+1} \\
 &= \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n} + \frac{1}{2n+1} + \frac{1}{2n+2} \\
 &= \frac{1}{n+1} - \frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n} + \frac{1}{2n+1} + \frac{1}{2n+2} \\
 &= \left(\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n} \right) + \frac{1}{2n+1} + \frac{1}{2n+2} - \frac{1}{n+1} \\
 &= y_n + \frac{1}{2n+1} + \frac{1}{2n+2} - \frac{1}{n+1} \\
 &\geq y_n + \frac{1}{2n+2} + \frac{1}{2n+2} - \frac{1}{n+1} \\
 &= y_n + \frac{1+1-2}{2n+2} \\
 &= y_n + 0 \\
 &= y_n
 \end{aligned}$$

Therefore, since $y_{n+1} \geq y_n$, $\forall n \in \mathbb{N}$, the sequence (y_n) is increasing.

Thus, since (y_n) is bounded above and increasing, by the Monotone Convergence Theorem, (y_n) converges.

Exercise 3.3.12 Establish the convergence and find the limit of the following sequences:

(a) $\left(1 + \frac{1}{n}\right)^{n+1}$.

Solution: